Saipranav Venkatakrishnan

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Education

University of Illinois Urbana-Champaign
B.S. in Computer Engineering — August 2022 – May 2026 (GPA: 3.85)
M.S./PhD in Computer Engineering — August 2026 – May 2032

Skills & Tools: SystemVerilog, Verilog, C/C++, CUDA C++, Python; Synopsys Design Compiler, VCS, Verdi, Verilator, Yosys, Spyglass Linter, CDC/RDC Analysis, Nvidia Nsight Compute/Systems, Git, PyTorch; ASIC & RTL Design, Hardware–Software Co-design, Computer Architecture, VLSI Design, Logic Synthesis, Parallel Programming

Experience

ASIC Design Intern — NVLink — May 2026 – August 2026
NVIDIA — Santa Clara, CA
- Designed **20+ production Verilog modules** for the **NVLink Chip-to-Chip interconnect**, implementing **packet framing/deframing**, **interconnect fanout reduction**, and **asynchronous/synchronous transaction** handling
- Implemented **CDC (Clock Domain Crossing)** logic for high-speed chip-to-chip links, applying synchronizer chains and handshake protocols to ensure **functional correctness** and **timing closure** across async clock boundaries
- Collaborated with **DV and Physical Design** teams to verify module integration, validate **CDC/RDC** constraints, and meet area and timing targets within the full NVLink SoC design flow

Undergraduate AI Hardware Research Assistant — December 2024 – Present
PlatformX — Champaign, IL
- Helped develop an **architectural simulator** for **AI wafer-scale accelerators**, enabling **hardware–software co-design** studies that identify improvements in **bandwidth utilization** and **compute efficiency** through reconfigurable designs
- Modeled **AI/ML workload dataflow** across accelerator subsystems to evaluate **performance–area–power tradeoffs**, providing architectural feedback to guide **compute and memory subsystem** design decisions
- Extended **GPGPU-Sim** (cycle-accurate simulator) with **C++** to enable detailed profiling of **memory access patterns** and **interconnect bandwidth**; applied across **GPU Processing Cluster (GPC)** designs studying **L1 cache slice** and **SM** interactions optimized for **AI workloads**

Computer Architecture (ECE 411) Course Staff — August 2025 – Present
UIUC Computer Organization and Design — Champaign, IL
- Trained students in designing and implementing **out-of-order RISC-V processors** with **non-blocking pipeline caching**, **branch prediction & recovery**, out-of-order **load/store units**, and **superscalar** execution
- Mentored students on **RTL debugging** across pipeline and cache projects using **Verdi** and **Verilator**; orchestrated competition profiling **30+ RISC-V processors** on **Power, Performance, and Area (PPA)** metrics

Graphics Engineering Intern — May 2025 – August 2025
Aechelon Technology — Roeland Park, KS
- Integrating SMPTE ST 2110-compliant **UDP transmission** into the pC-Nova platform via NVIDIA Rivermax SDK, utilizing **GPUDirect and RDMA** with full kernel bypass to eliminate **CPU bottlenecks**
- Implemented 5 **GPU kernels** and custom **fragment shaders** to transform output media into various color (e.g., YUV, sRGB) and vector spaces for compatibility with heterogeneous display systems
- Optimizing real-time media streaming by collaborating with senior engineers to leverage hardware-accelerated networking and GPU-based processing

Software Development Intern — May 2024–November 2024
CME Group — Chicago, IL
- Lead a team of 4 interns in developing an application to **monitor project deployments** to on-premises and cloud server for any **abnormal resource allocations** and stalls
- Won **first place** in an internal **CME Hackathon** to develop a **low-latency trading algorithm**, earning $1000 as the prize

Software Development Intern — March 2023–August 2023
CME Group — Champaign, IL
- Supervised a team of 5 intern through **creating** an internal **C++ library** to validate research papers about methods to foretell stock prices through corresponding future market data
- Configured a CME security compliant remote development environment and integrated over **3000+ lines** into the CME codebase

Projects

FORTE: Fault-Tolerant RISC-V Processor Tape-Out April 2025 – Present
Silicon Research Project (65nm Tape-Out) SystemVerilog, VCS Design Compiler, Verdi, Spyglass, Yosys

- Designed and implemented **fault-tolerance RTL** on a **CORE-V Wally** (RV64IMA, Sv39 MMU, PMP, M/S-mode) baseline targeting **65nm tape-out**; features include **SECDED ECC** across register file, L1 I/D-cache, PTW/TLB, a **cacheline and CSR scrubbing FSM**, and a **shadow pipeline** for redundant execution-unit fault detection
- Implemented **clock integrity monitor** (reference-oscillator comparison, ±5% glitch threshold, pipeline flush/reset response) and **micro-TMR hardening** for security-critical CSRs (`mstatus`, `mepc`, `mtvec`, PMP regs) to defend against physical fault-injection attacks
- Built complete **hardware Root of Trust** infrastructure: boot ROM with **SHA-256/RSA-2048** image authentication, **TRNG** via prime-length ring oscillators (Zkr), and full Keystone **TEE co-design** (16 PMP CSRs, architectural state scrubbing on enclave entry/exit, debug port lockdown)
- Conducted area feasibility study using **VCS Design Compiler** using TSMC's 65nm PDK; designed **AXI-Lite chip-to-board interface** (~50 pins) for FPGA-based bringup

RV-32IM Out-of-Order RISC-V Processor January 2025 – May 2025
Computer Architecture Project SystemVerilog, C++

- Designed and verified a pipelined, out-of-order, **N-way superscalar processor** based on the **RV32IM ISA**, with explicit register renaming
- Implemented a split Load/Store Unit for **speculative memory operations**, branch predictors, and **non-blocking pipelined cache**, including **functional coverage** metrics to assess verification completeness
- Debugged RTL with waveform tracing using **Verilator** and **Verdi**; verification involved writing unit-level **SystemVerilog testbenches** and assertions to validate pipeline stages, and simulations were performed with industry tools (VCS-compatible)
- Performed synthesis and power-performance-timing (**PPA**) analysis using **Synopsys Design Compiler**; evaluated across multiple testcases for critical path delays, area constraints, and **dynamic power** efficiency

Graphics Engine May 2024 – August 2024
Graphics Project C++, Vulkan

- Building a **graphics pipeline** from scratch using the **Vulkan SDK**, with over **6K lines of C++ code** written across the entire codebase for low-latency rendering
- Implemented 5+ core **graphics pipeline stages**, including input assembly, rasterization, fragment shading, and framebuffer presentation
- Developed a system to automate Vulkan pipeline creation and manage 5+ **shader modules** across multiple **render passes**

Spiking Neural Network October 2024 – January 2025
Hardware Synthesis and Neuromorphic Computing SystemVerilog, FPGA

- Researched **neuromorphic architectures** and deployed a **spiking neural network** on an FPGA trained on the **Berkeley DROID Dataset** for Robotic Manipulation
- Trained Leaky-Integrate-and-Fire (LIF) neuron weights in **PyTorch**, performed **quantization** for hardware compatibility, and mapped parameters to fixed-point representations
- Designed LIF neuron modules using shift-register counters to emulate membrane potential dynamics and developed inter-layer **FIFO queues** to stage **asynchronous spike traffic**
- Streamed multi-gigabyte testing data through the **UART protocol** to the **FPGA** and back, testing inference accuracy against PyTorch model

Audio Visualizer Theremin October 2024 – December 2024
Hardware Synthesis and Verification SystemVerilog, FPGA

- Designed and implemented an FPGA-based SoC on Xilinx for real-time audio synthesis and VGA visualization, controlled via UART input; integrated a MicroBlaze soft processor with custom hardware modules (audio generator, VGA controller) for real-time processing
- Developed a modular SystemVerilog testbench framework for both component and SoC-level validation, featuring automated stimulus generation, self-checking monitors, and coverage tracking
- Validated and debugged designs in Vivado using assertions and waveform analysis to ensure correctness across audio processing states

32-bit Operating System January 2024 – May 2024
Systems Project C, x86, QEMU

- Programmed a 32-bit Operating System with a **3-terminal interface** for **IA-32** architecture using **x86 assembly and C**
- Developed a 2-Radix **Paging System** and **Allocator** for both 4KB and 4MB page
- Constructed a simple **virtual file-system** and over 10 **system call handlers** to interface with the kernel and the physical file-system
- Designed a **Round Robin Scheduling algorithm** for switching between various userspace programs

Convolutional Neural Network Optimization January 2024 – May 2024
Parallel Algorithm and Profiling Project CUDA, C++

- Researched and implemented a scalable **CNN forward pass layer** using **CUDA**, and verified optimization efficiency through the use of Nvidia Nsight Systems and Compute
- Optimized convolutions through the use of streams, **shared memory**, constant memory, matrix unrolling, **tensor cores**, and using custom **half2** data types

Pytorch Contributions
AI/ML Contributions

- Contributed to PyTorch by writing documentation during PyTorch annual Docathon
- Connected with PyTorch developers to understand code and write correct and detailed comments about their purpose

Student Organizations

Engineering Tutor January 2024–Present
Eta Kappa Nu Honor Society Champaign, IL

- Inducted as a **top 20% student** in the department; tutor peers in topics ranging from LC-3 **assembly** and **C** programming to analog **signal processing**, **multivariable calculus**, and **architecture design**
- Create custom study guides and **host review sessions** across 5+ engineering courses, supporting **100+ students** in exam preparation

SIGArch Champaign, IL
Member January 2023 – Present

- Facilitated **technical discussions and reading groups** on computer architecture topics, including modern **CPUs**, **GPUs**, and **emerging accelerator designs**
- Helped coordinate architecture-focused talks and workshops featuring graduate students and faculty working on **microarchitecture, memory systems, and parallel computing**
- Collaborated with other **computing organizations** to host events bridging architecture with **systems, compilers, and hardware-software co-design**
- Supported **professional and academic networking events** connecting undergraduate students with graduate researchers and industry professionals in computer architecture

Social Chair May 2023 – May 2024
Association for Computing Machinery Champaign, IL

- Organized and hosted **50+ social events**, including weekly social hours that connected engineering students from freshmen to graduate levels
- **Expanded organization membership** by over 300 students through active community engagement and **cross-disciplinary outreach**
- Collaborated with **4+ campus organizations** to **plan minority-inclusive events** and co-host a large-scale joint school dance
- Assisted in coordinating **12+ professional networking events**, enabling students to engage with **industry professionals** across diverse computing disciplines